

Supplementary appendix

From Composite to Components: Evaluating the Impact of Domain Differentiation on Bayley-III Cognitive Diagnostic Precision

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Supplementary appendix to the manuscript submitted to The Lancet Child & Adolescent Health. Tables S1–S8 and Figures S1–S8; appendix references are numbered separately from those of the main text. In the tables, .. marks a value that does not apply or was not given.

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Abbreviations

ATT Attention · **WM** Working Memory · **GDPS** Goal-Directed Problem Solving · **FS** Flexibility/Shift · **HOP** Higher-Order Processing. ARI adjusted Rand index; NMI normalised mutual information; CFA confirmatory factor analysis; MIMIC multiple indicators multiple causes; WLSMV weighted least squares, mean- and variance-adjusted; STDYX/STDY Mplus standardisations (fully / semi-standardised); UC unclassified cluster; dHCP developing Human Connectome Project.

Supplementary tables

Table S1. Bayley-4 → Bayley-III translation cascade

The 39 Bayley-4 → Bayley-III item pairs produced by the translation cascade, with full-text and title cosine similarities and the stage at which each pair was locked (27 title-locked, 12 full-text fallback).

Bayley-4 item	Bayley-4 title	Bayley-III item	Bayley-III title	Aylward domain(s)	Transferred domain	Cosine (full text)	Cosine (title)	Stage
COG_002	Looks at Object	COG_003	Regards Object for 3 Seconds	ATT	ATT	0.891	0.4	full-text fallback
COG_003	Habituates to Rattle	COG_004	Habituates to Rattle	ATT	ATT	0.983	1.0	title-locked
COG_008	Shifts Attention	COG_010	Shifts Attention	ATT	ATT	0.96	1.0	title-locked
COG_009	Shows Visual Preference	COG_011	Shows Visual Preference	ATT	ATT	0.933	1.0	title-locked
COG_010	Habituates to Balloons	COG_014	Habituates to Picture (Balloons)	ATT	ATT	0.909	0.83	title-locked
COG_011	Prefers Ball	COG_015	Prefers Novel Picture (Ball)	ATT	ATT	0.833	0.704	full-text fallback
COG_012	Responds to Surroundings	COG_002	Responds to Surroundings Series: Inspects	ATT	ATT	0.91	0.598	full-text fallback
COG_026	Looks at Pictures	COG_032	Looks at Pictures	ATT	ATT	0.973	1.0	title-locked
COG_041	Listens to Story	COG_059	Attends to Story	ATT	ATT	0.887	0.839	title-locked
COG_034	Removes Pellet	COG_042	Removes Pellet	FS	FS	0.919	1.0	title-locked
COG_040	Pink Board Series: 2 Pieces	COG_049	Pink Board Series: 1 Piece	FS	FS	0.963	0.959	title-locked
COG_043	Blue Board Series: 1 Piece	COG_051	Blue Board Series: 1 Piece	FS	FS	0.977	1.0	title-locked
COG_049	Blue Board Series: 4 Pieces	COG_058	Blue Board Series: 4 Pieces	FS	FS	0.974	1.0	title-locked
COG_051	Rotated Pink Board	COG_060	Rotated Pink Board	FS + HOP	FS	0.958	1.0	title-locked
COG_052	Object Assembly: ball	COG_061	Object Assembly (Ball)	FS	FS	0.934	0.976	title-locked
COG_053	Object Assembly: Ice Cream Cone	COG_063	Object Assembly (Ice Cream Cone)	FS	FS	0.953	0.99	title-locked
COG_073	Object Assembly: Dog	COG_082	Object Assembly (Dog)	FS	FS	0.946	0.972	title-locked
COG_074	Object Assembly: Cat	COG_066	Blue Board Series: Completes (75 Seconds)	FS	FS	0.625	-0.025	full-text fallback
COG_014	Brings to Mouth	COG_017	Carries Object to Mouth	GDPS	GDPS	0.867	0.556	full-text fallback
COG_022	Obtains Ring	COG_041	Suspends Ring	GDPS	GDPS	0.902	0.51	full-text fallback
COG_035	Clear Box	COG_043	Clear Box: Front	GDPS + FS	GDPS	0.775	0.796	title-locked
COG_048	Uses Pencil to Obtain Object	COG_057	Uses Pencil to Obtain Object	GDPS + HOP	GDPS	0.96	1.0	title-locked

Bayley-4 item	Bayley-4 title	Bayley-III item	Bayley-III title	Aylward domain(s)	Transferred domain	Cosine (full text)	Cosine (title)	Stage
COG_057	Imitates 2-Step Action	COG_067	Imitates a Two-Step Action	GDPS	GDPS	0.96	0.981	title-locked
COG_062	Grouping: Colour	COG_072	Concept Grouping: Color	HOP	HOP	0.564	0.828	title-locked
COG_063	Grouping: Size	COG_073	Concept Grouping: Size	HOP	HOP	0.567	0.776	title-locked
COG_068	Simple Pattern	COG_077	Simple Pattern	HOP	HOP	0.532	1.0	title-locked
COG_081	Completes Patterns	COG_091	Completes Patterns	HOP	HOP	0.958	1.0	title-locked
COG_021	Searches for Object	COG_025	Searches for Fallen Object	WM	WM	0.964	0.707	full-text fallback
COG_024	Anticipatory Gaze	COG_013	Prefers Novel Object	WM	WM	0.703	0.17	full-text fallback
COG_028	Searches for Missing Objects	COG_034	Searches for Missing Objects	WM	WM	0.975	1.0	title-locked
COG_032	Finds Hidden Object	COG_040	Finds Hidden Object	WM	WM	0.755	1.0	title-locked
COG_037	Finds Hidden Object (Reversed)	COG_045	Finds Hidden Object (Reversed)	WM	WM	0.762	1.0	title-locked
COG_042	Finds Hidden Object (Visible Displacement)	COG_050	Finds Hidden Object (Visible Displacement)	WM	WM	0.775	1.0	title-locked
COG_056	Spatial Memory: 3 Cards	COG_068	Matches 3 Colors	WM	WM	0.582	0.132	full-text fallback
COG_060	Recalls Names	COG_081	Identifies 3 Incomplete Pictures	WM	WM	0.66	0.162	full-text fallback
COG_064	Repeats Words	COG_064	Matches Pictures	WM	WM	0.553	0.155	full-text fallback
COG_067	Discriminates Pictures	COG_076	Discriminates Pictures	WM	WM	0.876	1.0	title-locked
COG_076	Spatial Memory: 6 Cards	COG_084	Spatial Memory	WM	WM	0.897	0.557	full-text fallback
COG_080	Repeats Number Sequences	COG_090	Repeats Number Sequences	WM	WM	0.89	1.0	title-locked

Aylward and colleagues assign 39 of the 81 Bayley-4 cognitive items to a domain. Three of these (COG_035, COG_048, COG_051) carry two domains and were transferred to the Bayley-III under their first-listed domain. One further item (COG_047) is assigned only in the empirically derived model; it was retained in the 40-item benchmark comparison (Objective 2) but has no expert domain to transfer. In concordance counts, multi-factor items were scored as concordant when the cluster matched either of their domains.

Table S2. Bayley-III cognitive items administered in the dHCP sample

The 35 items in the administered range (34–68), their assigned factor, and their status in the analysis: 23 retained, 11 excluded for floor or ceiling effects ($\geq 90\%$ identical responses among valid administrations), and 1 dropped as the sole remaining indicator of a factor.

Item	Title	Domain (translated)	Empirical factor	N valid	% pass	% fail	Dominant %	Status
COG034	Searches for Missing Objects	WM	WM	739	98.2	1.8	98.2	ceiling
COG035	Takes Blocks Out of Cup	FS	FS	739	97.8	2.2	97.8	ceiling
COG036	Block Series: 1 Block	FS	FS	739	98.5	1.5	98.5	ceiling
COG037	Picks Up Block Series: 3 Blocks	GDPS	GDPS	739	90.1	9.9	90.1	ceiling
COG038	Explores Holes in Pegboard	GDPS	GDPS	739	97.6	2.4	97.6	ceiling

Item	Title	Domain (translated)	Empirical factor	N valid	% pass	% fail	Dominant %	Status
COG039	Pushes Car	GDPS	GDPS	739	97·6	2·4	97·6	ceiling
COG040	Finds Hidden Object	WM	WM	739	87·3	12·7	87·3	retained
COG041	Suspends Ring	GDPS	GDPS	739	72·0	28·0	72·0	retained
COG042	Removes Pellet	FS	FS	739	96·6	3·4	96·6	ceiling
COG043	Clear Box: Front	GDPS	GDPS	739	85·7	14·3	85·7	retained
COG044	Squeezes Object	GDPS	GDPS	739	88·8	11·2	88·8	retained
COG045	Finds Hidden Object (Reversed)	WM	WM	739	62·8	37·2	62·8	retained
COG046	Removes Lid From Bottle	GDPS	GDPS	739	48·7	51·3	51·3	retained
COG047	Pegboard Series: 2 Holes	FS	FS	739	96·5	3·5	96·5	ceiling
COG048	Relational Play Series: Self	GDPS	GDPS	739	89·2	10·8	89·2	retained
COG049	Pink Board Series: 1 Piece	FS	FS	739	84·4	15·6	84·4	retained
COG050	Finds Hidden Object (Visible Displacement)	WM	WM	739	72·7	27·3	72·7	retained
COG051	Blue Board Series: 1 Piece	FS	FS	739	88·0	12·0	88·0	retained
COG052	Clear Box: Sides	GDPS	GDPS	739	68·3	31·7	68·3	retained
COG053	Relational Play Series: Others	GDPS	GDPS	739	66·6	33·4	66·6	retained
COG054	Block Series: 9 Blocks	FS	FS	739	66·8	33·2	66·8	retained
COG055	Pegboard Series: 6 Pegs	FS	FS	739	87·8	12·2	87·8	retained
COG056	Pink Board Series: Completes	FS	FS	739	55·9	44·1	55·9	retained
COG057	Uses Pencil to Obtain Object	GDPS	GDPS	739	18·4	81·6	81·6	retained
COG058	Blue Board Series: 4 Pieces	FS	FS	739	56·6	43·4	56·6	retained
COG059	Attends to Story	ATT	..	..	..	..	..	excluded: domain with a single indicator
COG060	Rotated Pink Board	FS	FS	739	38·8	61·2	61·2	retained
COG061	Object Assembly (Ball)	FS	FS	739	24·1	75·9	75·9	retained
COG062	Completes Pegboard: 25 Seconds	FS	FS	739	15·0	85·0	85·0	retained
COG063	Object Assembly (Ice Cream Cone)	FS	FS	739	11·4	88·6	88·6	retained
COG064	Matches Pictures	WM	WM	739	23·7	76·3	76·3	retained
COG065	Representational Play	FS	GDPS	739	3·1	96·9	96·9	floor
COG066	Blue Board Series: Completes (75 Seconds)	FS	FS	739	17·6	82·4	82·4	retained
COG067	Imitates a Two- Step Action	GDPS	GDPS	739	2·7	97·3	97·3	floor
COG068	Matches 3 Colors	WM	WM	739	9·3	90·7	90·7	floor

COG059 carries no response statistics because it was dropped before estimation: as the only attention item in the administered range it would have been the single indicator of its factor, which is unidentifiable.

Table S3. Confirmatory factor analysis

Confirmatory factor analysis of the 23 retained Bayley-III cognitive items: standardised (STDYX) loadings and inter-factor correlations for the 3-factor model, model fit statistics, and loadings for the unidimensional comparison model.

S3a: 3-factor model: standardised loadings (STDYX)

Factor	Item	Title	Loading	SE	Est./SE	p
WM	COG040	Finds Hidden Object	0.436	0.063	6.876	<0.0001
WM	COG045	Finds Hidden Object (Reversed)	0.517	0.045	11.602	<0.0001
WM	COG050	Finds Hidden Object (Visible Displacement)	0.614	0.045	13.654	<0.0001
WM	COG064	Matches Pictures	0.976	0.051	19.32	<0.0001
GDPS	COG041	Suspends Ring	0.461	0.054	8.562	<0.0001
GDPS	COG043	Clear Box: Front	0.621	0.062	10.006	<0.0001
GDPS	COG044	Squeezes Object	0.577	0.066	8.726	<0.0001
GDPS	COG046	Removes Lid From Bottle	0.41	0.052	7.882	<0.0001
GDPS	COG048	Relational Play Series: Self	0.833	0.051	16.188	<0.0001
GDPS	COG052	Clear Box: Sides	0.594	0.046	12.87	<0.0001
GDPS	COG053	Relational Play Series: Others	0.699	0.038	18.167	<0.0001
GDPS	COG057	Uses Pencil to Obtain Object	0.485	0.063	7.705	<0.0001
FS	COG049	Pink Board Series: 1 Piece	0.829	0.03	27.476	<0.0001
FS	COG051	Blue Board Series: 1 Piece	0.799	0.041	19.499	<0.0001
FS	COG054	Block Series: 9 Blocks	0.433	0.044	9.939	<0.0001
FS	COG055	Pegboard Series: 6 Pegs	0.836	0.038	22.235	<0.0001
FS	COG056	Pink Board Series: Completes	0.93	0.015	62.403	<0.0001
FS	COG058	Blue Board Series: 4 Pieces	0.82	0.022	37.278	<0.0001
FS	COG060	Rotated Pink Board	0.909	0.015	59.842	<0.0001
FS	COG061	Object Assembly (Ball)	0.815	0.025	32.052	<0.0001
FS	COG062	Completes Pegboard: 25 Seconds	0.649	0.042	15.425	<0.0001
FS	COG063	Object Assembly (Ice Cream Cone)	0.835	0.031	26.956	<0.0001
FS	COG066	Blue Board Series: Completes (75 Seconds)	0.952	0.019	50.293	<0.0001

S3b: 3-factor model: inter-factor correlations

Factor pair	Correlation	SE	Est./SE	p
WM-GDPS	0.616	0.05	12.402	<0.0001
WM-FS	0.705	0.041	17.259	<0.0001
GDPS-FS	0.54	0.037	14.394	<0.0001

S3c: model fit statistics and nested-model test

Statistic	3-factor (WM, GDPS, FS)	Unidimensional (COG)
Items	23	23
χ^2	820.91	1193.599
df	227	230
p (χ^2)	<0.0001	<0.0001
RMSEA	0.06	0.075
CFI	0.929	0.885
TLI	0.921	0.874
SRMR	0.108	0.113
90% CI low	0.055	0.071

Statistic	3-factor (WM, GDPS, FS)	Unidimensional (COG)
90% CI high	0.064	0.08
$\Delta\chi^2$ vs 3-factor	..	246.109
Δdf	..	3
p ($\Delta\chi^2$)	..	<0.0001

S3d: unidimensional model: standardised loadings (STDYX)

Item	Title	Loading	SE	Est./SE	p
COG040	Finds Hidden Object	0.328	0.058	5.683	<0.0001
COG045	Finds Hidden Object (Reversed)	0.401	0.042	9.616	<0.0001
COG050	Finds Hidden Object (Visible Displacement)	0.483	0.042	11.617	<0.0001
COG064	Matches Pictures	0.749	0.029	25.455	<0.0001
COG041	Suspends Ring	0.35	0.045	7.758	<0.0001
COG043	Clear Box: Front	0.475	0.053	8.996	<0.0001
COG044	Squeezes Object	0.436	0.058	7.559	<0.0001
COG046	Removes Lid From Bottle	0.329	0.043	7.67	<0.0001
COG048	Relational Play Series: Self	0.551	0.054	10.151	<0.0001
COG052	Clear Box: Sides	0.447	0.041	10.943	<0.0001
COG053	Relational Play Series: Others	0.465	0.039	11.983	<0.0001
COG057	Uses Pencil to Obtain Object	0.387	0.05	7.784	<0.0001
COG049	Pink Board Series: 1 Piece	0.81	0.03	26.869	<0.0001
COG051	Blue Board Series: 1 Piece	0.776	0.041	19.08	<0.0001
COG054	Block Series: 9 Blocks	0.429	0.042	10.184	<0.0001
COG055	Pegboard Series: 6 Pegs	0.819	0.037	22.382	<0.0001
COG056	Pink Board Series: Completes	0.92	0.015	61.657	<0.0001
COG058	Blue Board Series: 4 Pieces	0.799	0.022	36.651	<0.0001
COG060	Rotated Pink Board	0.902	0.015	59.782	<0.0001
COG061	Object Assembly (Ball)	0.803	0.025	32.158	<0.0001
COG062	Completes Pegboard: 25 Seconds	0.638	0.041	15.509	<0.0001
COG063	Object Assembly (Ice Cream Cone)	0.822	0.031	26.688	<0.0001
COG066	Blue Board Series: Completes (75 Seconds)	0.933	0.019	49.816	<0.0001

Nested-model test (S3c): WLSMV-corrected chi-square difference test (Mplus DIFFTEST) of the unidimensional model, in which the three factor correlations are fixed to 1, against the three-factor model on the same 23 items: $\Delta\chi^2(3) = 246.1$, $p < 0.0001$.

Table S4. MIMIC covariate associations

MIMIC models: standardised associations between 18 perinatal, neonatal and sociodemographic covariates and the latent factors (3-factor model) or the unidimensional composite. Continuous covariates are fully standardised (STDYX) and binary covariates semi-standardised (STDY), the convention used in the text and Figure 4A; the standardisation is stated per row. S4b gives the WLSMV chi-square difference tests (Mplus DIFFTEST) of the factor contrasts quoted in the text.

Covariate instruments. Maternal postnatal depression was assessed with the 10-item Edinburgh Postnatal Depression Scale (EPDS),¹ a maternal self-report. Dysfunctional parenting was assessed with the 30-item Parenting Scale,² which yields overreactivity, laxness, and verbosity subscales and a total score. Cognitively stimulating parenting was assessed with the Cognitively Stimulating Parenting Scale (CSPS).³

S4a: standardised regression coefficients

Model	Factor	Standardisation	Estimate	SE	z	p Sig.
AntCor Complete						

Model	Factor	Standardisation	Estimate	SE	z	p Sig.
3F	FS	STDY	-0.276	0.124	-2.231	0.026 *
3F	GDPS	STDY	-0.171	0.129	-1.329	0.18
3F	WM	STDY	-0.019	0.142	-0.133	0.89
UNIDIM	COG (unidimensional)	STDY	-0.242	0.121	-1.995	0.046 *
AntCor Partial						
3F	FS	STDY	-0.407	0.233	-1.748	0.08 †
3F	GDPS	STDY	-0.432	0.26	-1.664	0.096 †
3F	WM	STDY	0.228	0.303	0.75	0.45
UNIDIM	COG (unidimensional)	STDY	-0.363	0.238	-1.521	0.13
Birth Wt (g)						
3F	FS	STDYX	0.139	0.042	3.349	0.001 **
3F	GDPS	STDYX	0.112	0.045	2.496	0.013 *
3F	WM	STDYX	0.073	0.051	1.434	0.15
UNIDIM	COG (unidimensional)	STDYX	0.138	0.04	3.416	0.001 **
Paternal Edu (y)						
3F	FS	STDYX	-0.046	0.042	-1.105	0.27
3F	GDPS	STDYX	-0.135	0.054	-2.503	0.012 *
3F	WM	STDYX	-0.042	0.057	-0.743	0.46
UNIDIM	COG (unidimensional)	STDYX	-0.065	0.042	-1.559	0.12
Postnatal depression (EPDS)						
3F	FS	STDYX	-0.08	0.043	-1.856	0.063 †
3F	GDPS	STDYX	-0.156	0.047	-3.309	0.001 **
3F	WM	STDYX	-0.1	0.052	-1.944	0.052 †
UNIDIM	COG (unidimensional)	STDYX	-0.107	0.043	-2.485	0.013 *
GA (wks)						
3F	FS	STDYX	0.159	0.041	3.824	<0.0001 ***
3F	GDPS	STDYX	0.111	0.043	2.588	0.01 *
3F	WM	STDYX	0.096	0.049	1.954	0.051 †
UNIDIM	COG (unidimensional)	STDYX	0.156	0.041	3.817	<0.0001 ***
Gest. Diabetes						
3F	FS	STDY	-0.202	0.174	-1.159	0.25
3F	GDPS	STDY	-0.028	0.215	-0.132	0.90
3F	WM	STDY	-0.296	0.239	-1.236	0.22
UNIDIM	COG (unidimensional)	STDY	-0.198	0.167	-1.19	0.23
Head Circ (cm)						
3F	FS	STDYX	0.179	0.044	4.05	<0.0001 ***
3F	GDPS	STDYX	0.118	0.045	2.599	0.009 **
3F	WM	STDYX	0.116	0.053	2.176	0.03 *
UNIDIM	COG (unidimensional)	STDYX	0.174	0.043	4.102	<0.0001 ***
IUGR (Yes vs No)						
3F	FS	STDY	-0.206	0.17	-1.214	0.23
3F	GDPS	STDY	0.063	0.211	0.297	0.77
3F	WM	STDY	-0.057	0.21	-0.271	0.79
UNIDIM	COG (unidimensional)	STDY	-0.148	0.178	-0.83	0.41
Maternal Age (y)						
3F	FS	STDYX	0.041	0.039	1.05	0.29
3F	GDPS	STDYX	-0.002	0.044	-0.039	0.97
3F	WM	STDYX	0.125	0.048	2.596	0.009 **
UNIDIM	COG (unidimensional)	STDYX	0.049	0.038	1.288	0.20
Maternal Edu (y)						
3F	FS	STDYX	0.001	0.041	0.018	0.99
3F	GDPS	STDYX	-0.072	0.048	-1.493	0.14

Model	Factor	Standardisation	Estimate	SE	z	p Sig.
3F	WM	STDYX	0.027	0.052	0.51	0.61
UNIDIM	COG (unidimensional)	STDYX	-0.009	0.042	-0.216	0.83
Maternal English						
3F	FS	STDY	0.198	0.083	2.378	0.017 *
3F	GDPS	STDY	0.251	0.092	2.718	0.007 **
3F	WM	STDY	0.309	0.102	3.03	0.002 **
UNIDIM	COG (unidimensional)	STDY	0.246	0.081	3.035	0.002 **
Pre-eclampsia						
3F	FS	STDY	-0.071	0.148	-0.484	0.63
3F	GDPS	STDY	-0.148	0.171	-0.865	0.39
3F	WM	STDY	-0.026	0.169	-0.156	0.88
UNIDIM	COG (unidimensional)	STDY	-0.091	0.152	-0.595	0.55
Parenting laxness						
3F	FS	STDYX	-0.034	0.043	-0.795	0.43
3F	GDPS	STDYX	-0.057	0.05	-1.151	0.25
3F	WM	STDYX	-0.024	0.055	-0.443	0.66
UNIDIM	COG (unidimensional)	STDYX	-0.04	0.044	-0.913	0.36
Parenting overreactivity						
3F	FS	STDYX	0.021	0.043	0.485	0.63
3F	GDPS	STDYX	-0.011	0.056	-0.202	0.84
3F	WM	STDYX	0.032	0.053	0.612	0.54
UNIDIM	COG (unidimensional)	STDYX	0.02	0.044	0.464	0.64
Parenting Style						
3F	FS	STDYX	-0.005	0.043	-0.116	0.91
3F	GDPS	STDYX	-0.068	0.052	-1.305	0.19
3F	WM	STDYX	0.01	0.054	0.179	0.86
UNIDIM	COG (unidimensional)	STDYX	-0.013	0.043	-0.297	0.77
Parenting verbosity						
3F	FS	STDYX	0.016	0.043	0.377	0.71
3F	GDPS	STDYX	-0.078	0.05	-1.567	0.12
3F	WM	STDYX	-0.039	0.053	-0.725	0.47
UNIDIM	COG (unidimensional)	STDYX	-0.008	0.044	-0.183	0.85
Sex (M vs F)						
3F	FS	STDY	-0.094	0.078	-1.206	0.23
3F	GDPS	STDY	-0.365	0.085	-4.274	<0.0001 ***
3F	WM	STDY	-0.241	0.092	-2.631	0.009 **
UNIDIM	COG (unidimensional)	STDY	-0.172	0.076	-2.257	0.024 *
Stimulating home environment						
3F	FS	STDYX	0.185	0.04	4.642	<0.0001 ***
3F	GDPS	STDYX	0.224	0.043	5.152	<0.0001 ***
3F	WM	STDYX	0.242	0.052	4.65	<0.0001 ***
UNIDIM	COG (unidimensional)	STDYX	0.221	0.038	5.753	<0.0001 ***

S4b: factor contrasts (WLSMV DIFFTEST, 1 df)

Covariate	Constraint (H0)	$\Delta\chi^2$	df	p Sig.
AntCor	AntCor_P: WM = GDPS	8.122	1	0.0044 **
AntCor	AntCor_C: WM = FS	6.798	1	0.0091 **
BirthWt	BirthWt: WM = FS	7.606	1	0.0058 **
DadEdu	DadEdu: WM = GDPS	2.533	1	0.11 ns
GA	GA: WM = FS	8.813	1	0.003 **
GDiab	GDiab: WM = GDPS	1.121	1	0.29 ns

Covariate	Constraint (H0)	$\Delta\chi^2$	df	p Sig.
HdCirc	HdCirc: WM = FS	8.406	1	0.0037 **
IUGR	IUGR: GDPS = FS	2.968	1	0.085 ns
MomAge	MomAge: WM = GDPS	5.982	1	0.015 *
MomEdu	MomEdu: WM = GDPS	3.521	1	0.061 ns
MomEng	MomEng: WM = FS	0.215	1	0.64 ns
PrEcl	PrEcl: WM = GDPS	0.597	1	0.44 ns
Sex	Sex: GDPS = FS	2.031	1	0.15 ns
StimEnv	StimEnv: WM = FS	2.114	1	0.15 ns
EPDSTot	epds: GDPS = FS	0.011	1	0.92 ns
PrLax	prlax: WM = GDPS	0.579	1	0.45 ns
PrOvr	provr: WM = GDPS	0.506	1	0.48 ns
PrVrb	prvrb: GDPS = FS	2.519	1	0.11 ns
PrSty	prsty: WM = FS	0.073	1	0.79 ns

† p < 0.1 · * p < 0.05 · ** p < 0.01 · *** p < 0.001

Table S5. Factor–outcome associations

Standardised latent correlations between the factors (3-factor model, and the unidimensional composite) and concurrent developmental and behavioural outcomes.

Model	Factor	Estimate	SE	z	p Sig.
Bayley Language · Bayley					
3F	FS	0.45	0.032	14.102	<0.0001 ***
3F	GDPS	0.533	0.03	17.684	<0.0001 ***
3F	WM	0.639	0.036	17.718	<0.0001 ***
UNIDIM	COG (unidimensional)	0.547	0.027	20.196	<0.0001 ***
Bayley Motor · Bayley					
3F	FS	0.409	0.031	13.01	<0.0001 ***
3F	GDPS	0.385	0.037	10.482	<0.0001 ***
3F	WM	0.444	0.044	10.093	<0.0001 ***
UNIDIM	COG (unidimensional)	0.451	0.029	15.462	<0.0001 ***
CBCL Externalising · CBCL Broad					
3F	FS	0.05	0.04	1.237	0.22
3F	GDPS	−0.02	0.045	−0.444	0.66
3F	WM	0.023	0.048	0.486	0.63
UNIDIM	COG (unidimensional)	0.037	0.04	0.924	0.35
CBCL Internalising · CBCL Broad					
3F	FS	−0.031	0.039	−0.799	0.42
3F	GDPS	−0.153	0.044	−3.491	<0.0001 ***
3F	WM	−0.119	0.047	−2.544	0.011 *
UNIDIM	COG (unidimensional)	−0.075	0.041	−1.838	0.066 †
CBCL Total Problems · CBCL Broad					
3F	FS	−0.005	0.04	−0.123	0.90
3F	GDPS	−0.102	0.044	−2.317	0.021 *
3F	WM	−0.076	0.047	−1.604	0.11
UNIDIM	COG (unidimensional)	−0.037	0.04	−0.905	0.37
CBCL Aggressive Behaviour · CBCL Narrow					
3F	FS	0.209	0.039	5.355	<0.0001 ***
3F	GDPS	0.121	0.041	2.968	0.003 **
3F	WM	0.218	0.045	4.849	<0.0001 ***
UNIDIM	COG (unidimensional)	0.215	0.037	5.839	<0.0001 ***
CBCL Anxious/Depressed · CBCL Narrow					
3F	FS	−0.069	0.038	−1.814	0.07 †
3F	GDPS	−0.119	0.044	−2.727	0.006 **
3F	WM	−0.09	0.046	−1.977	0.048 *

Model	Factor	Estimate	SE	z	p Sig.
UNIDIM	COG (unidimensional)	-0.088	0.039	-2.268	0.023 *
CBCL Attention Problems · CBCL Narrow					
3F	FS	-0.154	0.038	-4.022	<0.0001 ***
3F	GDPS	-0.135	0.044	-3.082	0.002 **
3F	WM	-0.227	0.047	-4.856	<0.0001 ***
UNIDIM	COG (unidimensional)	-0.177	0.038	-4.662	<0.0001 ***
CBCL Emotionally Reactive · CBCL Narrow					
3F	FS	0.042	0.039	1.075	0.28
3F	GDPS	-0.048	0.044	-1.084	0.28
3F	WM	-0.011	0.047	-0.234	0.81
UNIDIM	COG (unidimensional)	0.02	0.039	0.521	0.60
CBCL Sleep Problems · CBCL Narrow					
3F	FS	0.019	0.04	0.478	0.63
3F	GDPS	-0.017	0.042	-0.408	0.68
3F	WM	0.009	0.047	0.197	0.84
UNIDIM	COG (unidimensional)	0.013	0.039	0.329	0.74
CBCL Somatic Complaints · CBCL Narrow					
3F	FS	0.019	0.039	0.503	0.61
3F	GDPS	-0.071	0.04	-1.751	0.08 †
3F	WM	-0.058	0.049	-1.2	0.23
UNIDIM	COG (unidimensional)	-0.009	0.038	-0.239	0.81
CBCL Withdrawn/Depressed · CBCL Narrow					
3F	FS	-0.101	0.041	-2.492	0.013 *
3F	GDPS	-0.233	0.045	-5.236	<0.0001 ***
3F	WM	-0.243	0.046	-5.236	<0.0001 ***
UNIDIM	COG (unidimensional)	-0.167	0.042	-3.961	<0.0001 ***
Effortful Control · Emotion Control					
3F	FS	-0.072	0.039	-1.827	0.068 †
3F	GDPS	-0.008	0.044	-0.185	0.85
3F	WM	-0.014	0.049	-0.296	0.77
UNIDIM	COG (unidimensional)	-0.058	0.039	-1.482	0.14
Negative Affect · Emotion Control					
3F	FS	-0.042	0.041	-1.03	0.30
3F	GDPS	-0.064	0.044	-1.468	0.14
3F	WM	-0.044	0.051	-0.876	0.38
UNIDIM	COG (unidimensional)	-0.05	0.041	-1.214	0.23
Surgency/Extraversion · Emotion Control					
3F	FS	-0.019	0.04	-0.478	0.63
3F	GDPS	0.017	0.043	0.407	0.68
3F	WM	0.005	0.049	0.106	0.92
UNIDIM	COG (unidimensional)	-0.01	0.039	-0.244	0.81
Q-CHAT Total · Q-CHAT					
3F	FS	-0.276	0.037	-7.552	<0.0001 ***
3F	GDPS	-0.39	0.034	-11.382	<0.0001 ***
3F	WM	-0.362	0.046	-7.81	<0.0001 ***
UNIDIM	COG (unidimensional)	-0.346	0.034	-10.262	<0.0001 ***

† $p < 0.1$ · * $p < 0.05$ · ** $p < 0.01$ · *** $p < 0.001$

Table S6. Per-domain recovery of the Aylward domains by the Bayley-4 clusters

Number of Bayley-4 items in each expert (Aylward) domain whose unsupervised cluster was labelled with that domain, under two scoring rules: any-factor (an item counts as recovered if the cluster matches any of its listed domains; the rule used in the text) and single-domain (the three multi-factor items excluded, so every item has one domain).

Domain	Items (any-factor)	Recovered	%	Items (single-domain)	Recovered	%
ATT	9	9	100·0	9	9	100·0
WM	12	5	41·7	12	5	41·7
GDPS	5	0	0·0	3	0	0·0
FS	10	9	90·0	8	7	87·5
HOP	6	1	16·7	4	0	0·0
All items	39	24	61·5	36	21	58·3

Under the any-factor rule the multi-factor items appear in both of their domains, so the domain rows sum to 42 memberships; the "All items" row counts each of the 39 expert-assigned items once (24 recovered, 61·5%).

Table S7. Pair-level position of the Bayley-4 clusters between the two reference models

Every pair of Bayley-4 items with a domain in both of Aylward's reference models is classified as consensus-apart (both models separate the items), consensus-together (both models join them) or disputed (the models disagree). Columns give three treatments of the multi-factor items: excluded (31 single-domain items), any-factor (two items are together under the expert model if they share any listed domain), and first-listed domain (the resolution used for the transfer to the Bayley-III). COG_047 (no expert domain) and the five items absent from the empirical model (COG_002, 003, 008, 052, 081) are excluded throughout.

	31 single-domain items	Any-factor (34 items)	First-listed domain (34 items)
Consensus-apart pairs			
Pairs	299	358	367
Kept apart by the clusters	287 (96%)	345 (96%)	352 (96%)
Consensus-together pairs			
Pairs	39	52	46
Kept together by the clusters	30 (77%)	40 (77%)	35 (76%)
Disputed pairs			
Pairs	127	151	148
Resolved toward the expert model	67	81	80
Resolved toward the empirical model	60	70	68
Pairs placed together by each model			
Expert model	108	137	122
Empirical model	97	118	118
Detected clusters	87	104	104

In the 31-item treatment the nine broken consensus-together pairs all involve COG_024 and COG_067, which the clusters assigned to the attention cluster; the clusters merge 87 pairs against 108 (expert) and 97 (empirical). All values from b4_pair_level_intermediacy.py.

Table S8. Expert-review ledger

Items whose unsupervised cluster disagreed with the reference structure, with the two experts' independent domain assignments (blind to the algorithmic labels) and their consensus. (a) The 16 Bayley-4 items discordant with the Aylward domain under the any-factor rule (Objective 2). (b) The 60 Bayley-III items whose bottom-up cluster (Objective 3) differs from the final translated structure; the algorithmic label is the Step 1 assignment before the expert overrides, the final label the one carried into the 91-item model. Where an expert listed more than one domain the first-listed one is shown, as used for the kappa statistics; .. marks a blank or no consensus.

S8a: Bayley-4: 16 discordant items

Item	Aylward domain	Detected cluster	Expert 1	Expert 2	Consensus
COG_021	WM	GDPS	ATT	FS	..
COG_024	WM	ATT	WM	ATT	..
COG_028	WM	GDPS	GDPS	FS	..
COG_032	WM	GDPS	WM	WM	WM
COG_037	WM	GDPS	HOP	WM	..
COG_042	WM	GDPS	HOP	WM	..
COG_067	WM	ATT	WM	WM	WM
COG_014	GDPS	HOP	FS	HOP	..

Item	Aylward domain	Detected cluster	Expert 1	Expert 2	Consensus
COG_022	GDPS	HOP	GDPS	GDPS	GDPS
COG_057	GDPS	HOP	HOP	GDPS	..
COG_034	FS	HOP	GDPS	GDPS	GDPS
COG_062	HOP	WM	HOP	HOP	HOP
COG_063	HOP	WM	HOP	HOP	HOP
COG_068	HOP	WM	HOP	HOP	HOP
COG_081	HOP	FS	HOP	GDPS	..
COG_047	..	FS	HOP	HOP	HOP

S8b: Bayley-III: 60 discordant items

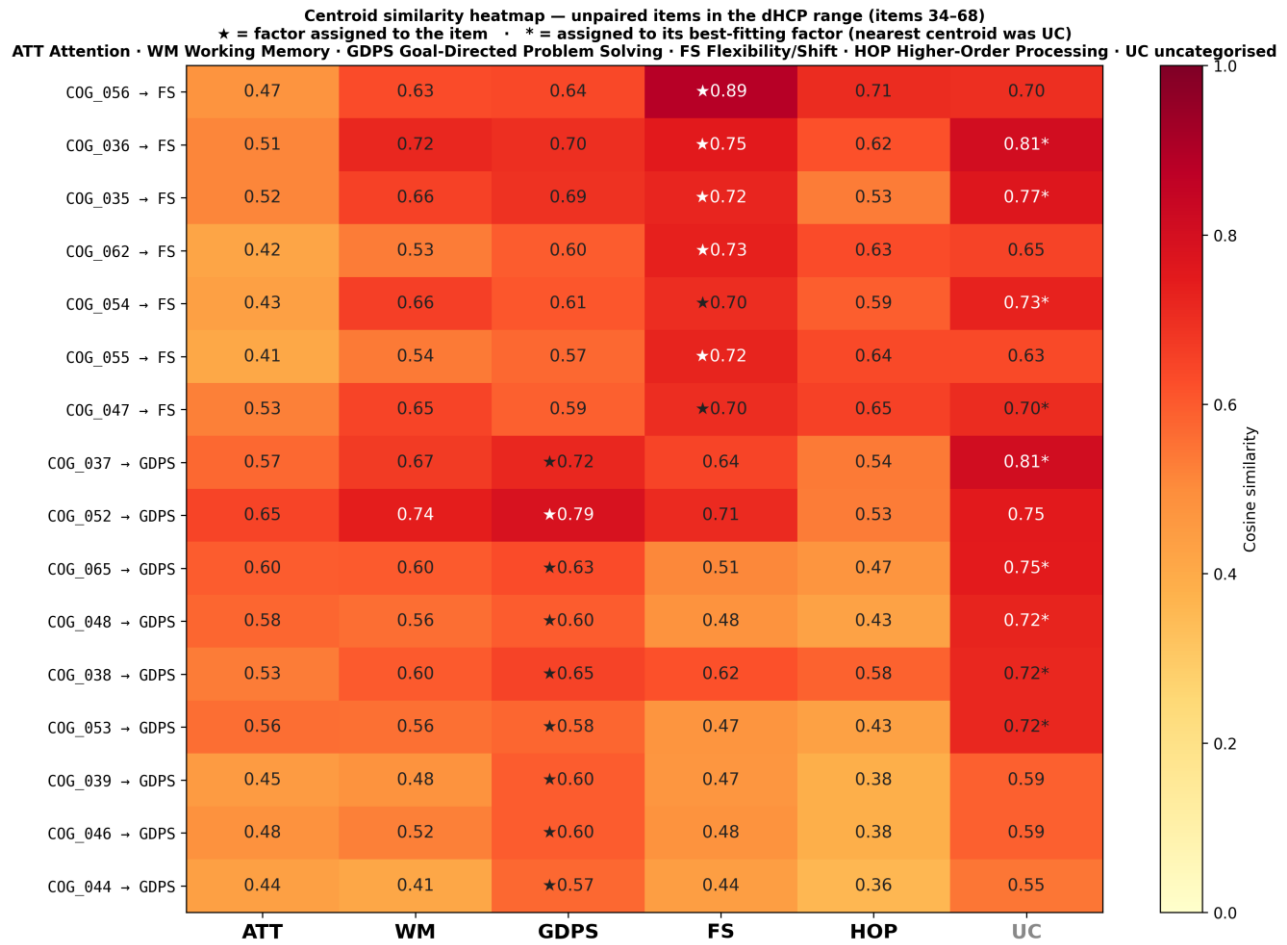
Item	Algorithmic label	Final label	Bottom-up cluster	Expert 1	Expert 2	Consensus
COG_011	ATT	ATT	WM	ATT	ATT	ATT
COG_015	ATT	ATT	WM	ATT	ATT	ATT
COG_016	ATT	ATT	FS	ATT	ATT	ATT
COG_033	GDPS	ATT	GDPS	ATT	ATT	ATT
COG_013	WM	WM	FS	WM	ATT	..
COG_025	WM	WM	FS	WM	ATT	..
COG_034	WM	WM	FS	WM	WM	WM
COG_040	WM	WM	FS	GDPS	WM	..
COG_045	WM	WM	FS	HOP	WM	..
COG_050	WM	WM	FS	WM	WM	WM
COG_079	WM	WM	GDPS	..	HOP	..
COG_085	WM	WM	GDPS	HOP	HOP	HOP
COG_086	WM	WM	GDPS	HOP	HOP	HOP
COG_089	WM	WM	GDPS	HOP	HOP	HOP
COG_090	WM	WM	GDPS	WM	WM	WM
COG_017	GDPS	GDPS	FS	..	ATT	..
COG_021	GDPS	GDPS	FS	GDPS	ATT	..
COG_023	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_024	GDPS	GDPS	FS	..	GDPS	..
COG_026	GDPS	GDPS	FS	..	GDPS	..
COG_028	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_029	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_031	GDPS	GDPS	FS	HOP	GDPS	..
COG_039	GDPS	GDPS	FS	..	GDPS	..
COG_041	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_043	GDPS	GDPS	FS	GDPS	ATT	..
COG_044	GDPS	GDPS	FS	..	ATT	..
COG_046	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_048	GDPS	GDPS	HOP	FS	GDPS	..
COG_052	GDPS	GDPS	FS	GDPS	ATT	..
COG_053	GDPS	GDPS	HOP	GDPS	GDPS	GDPS
COG_057	GDPS	GDPS	FS	GDPS	GDPS	GDPS
COG_067	GDPS	GDPS	HOP	WM	WM	WM
COG_071	GDPS	GDPS	HOP	FS	FS	FS
COG_030	FS	FS	GDPS	ATT	ATT	ATT
COG_035	FS	FS	GDPS	..	WM	..
COG_036	FS	FS	GDPS	FS	GDPS	..
COG_047	FS	FS	GDPS	GDPS	HOP	..
COG_049	FS	FS	GDPS	GDPS	HOP	..
COG_051	FS	FS	GDPS	GDPS	HOP	..
COG_054	FS	FS	GDPS	GDPS	GDPS	GDPS
COG_055	FS	FS	GDPS	GDPS	HOP	..
COG_056	FS	FS	GDPS	GDPS	HOP	..
COG_058	FS	FS	GDPS	GDPS	HOP	..
COG_060	FS	FS	GDPS	FS	HOP	..
COG_061	FS	FS	GDPS	WM	HOP	..
COG_062	FS	FS	GDPS	FS	HOP	..
COG_063	FS	FS	GDPS	WM	HOP	..
COG_065	GDPS	FS	HOP	FS	FS	FS
COG_066	FS	FS	GDPS	GDPS	HOP	..
COG_069	GDPS	FS	HOP	FS	FS	FS
COG_070	FS	FS	GDPS	HOP	FS	..

Item	Algorithmic label	Final label	Bottom-up cluster	Expert 1	Expert 2	Consensus
COG_082	FS	FS	GDPS	WM	HOP	..
COG_072	HOP	HOP	WM	ATT	FS	..
COG_073	HOP	HOP	WM	ATT	FS	..
COG_074	HOP	HOP	WM	FS	FS	FS
COG_075	HOP	HOP	WM	ATT	FS	..
COG_077	HOP	HOP	WM	ATT	FS	..
COG_078	HOP	HOP	GDPS	HOP	FS	..
COG_091	HOP	HOP	GDPS	HOP	FS	..

Expert assignments were scored against the algorithmic labels prior to the three expert overrides (κ vs translated 0.27, 0.19); against the final structure, which incorporates those overrides, they are 0.35, 0.25. Of the six UC-flagged items reviewed under Objective 1, COG_027 (reviewed but concordant) is excluded; the other five are discordant under the final structure and are listed in S8b, three of them (COG_033, COG_065, COG_069) carrying the consensus override as their final label. COG_048 (Bayley-4) was on the review sheet but is concordant under the any-factor rule and was not rated.

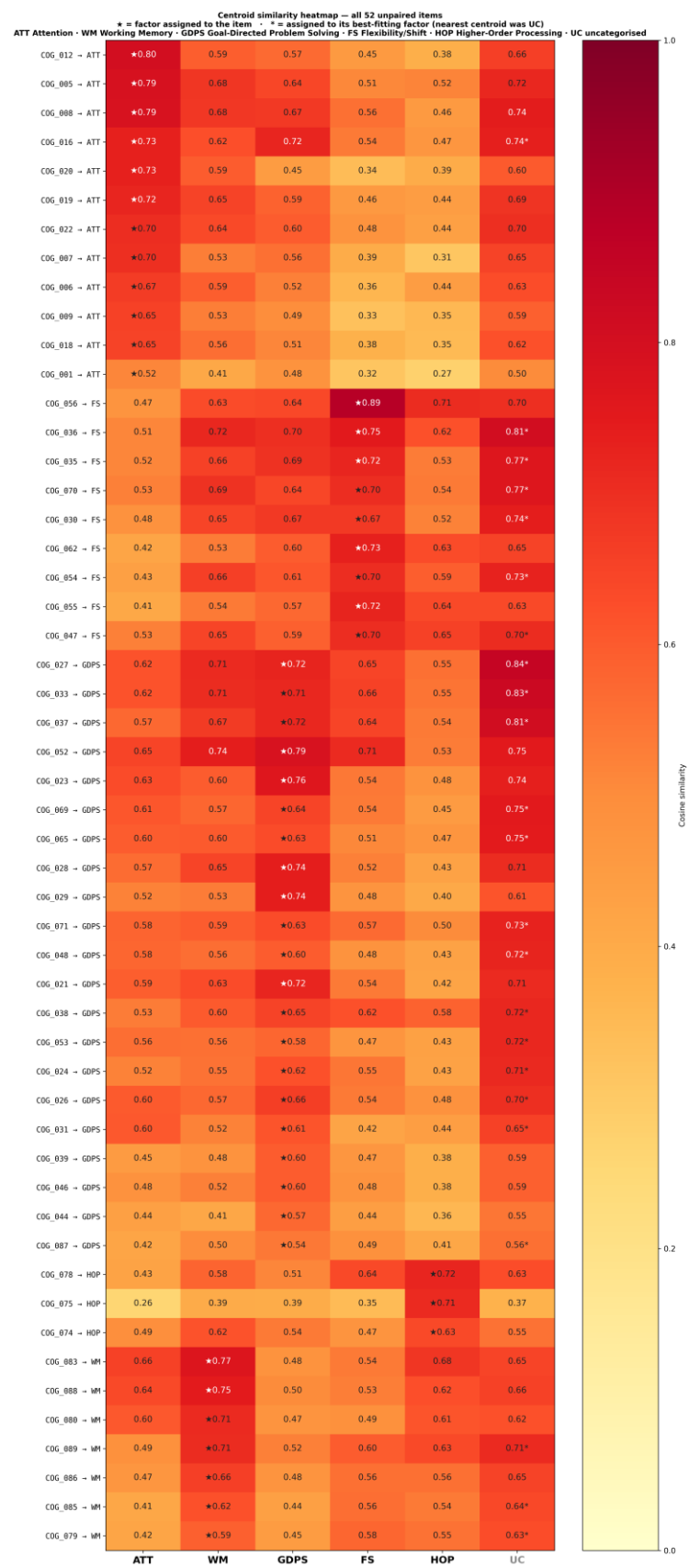
Supplementary figures

Figure S1. Centroid similarity profiles, unpaired items in the dHCP administration range



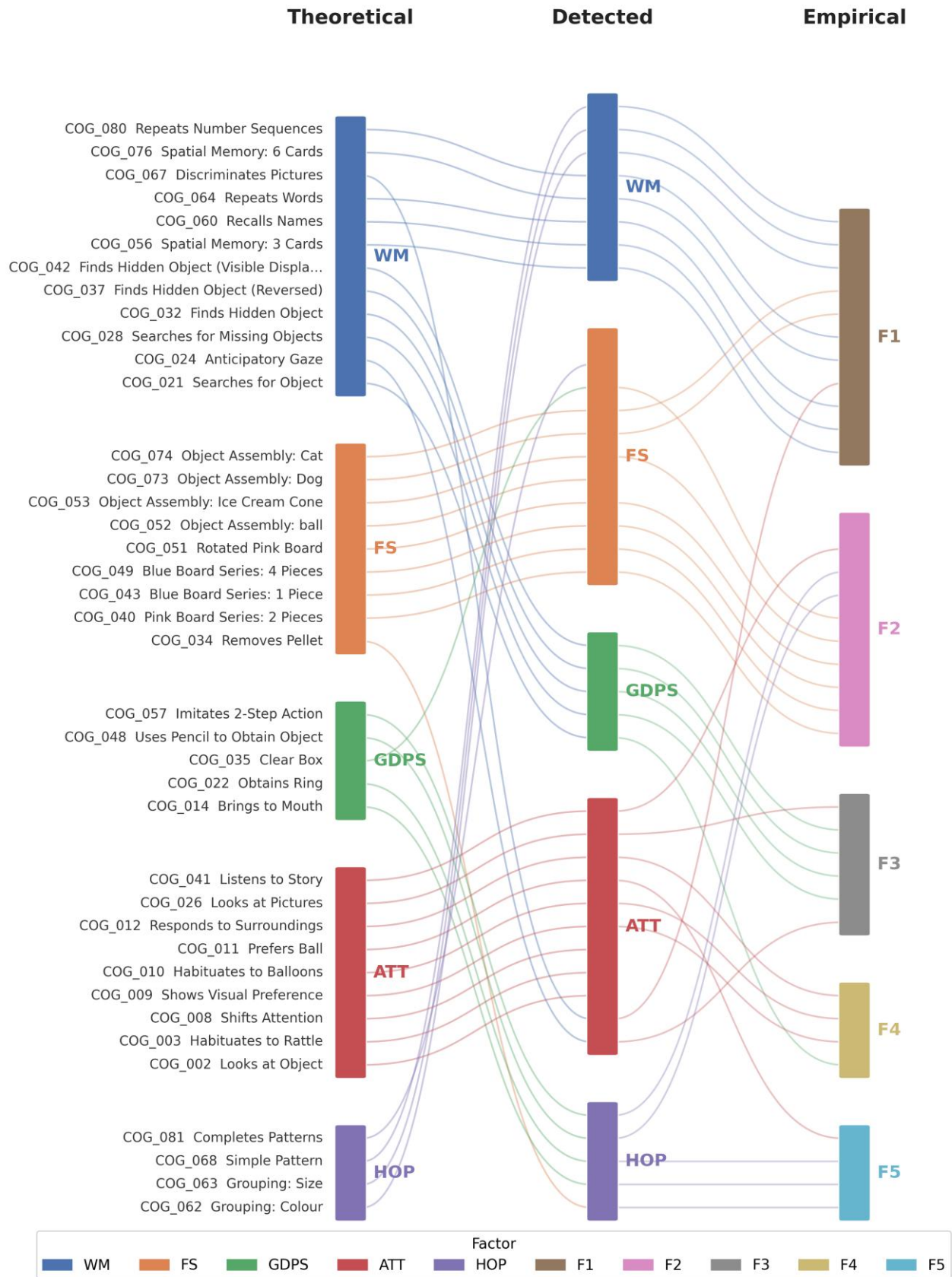
Centroid similarity heatmap for unpaired items in the dHCP administration range (items 34–68).

Figure S2. Centroid similarity profiles, all 52 unpaired items



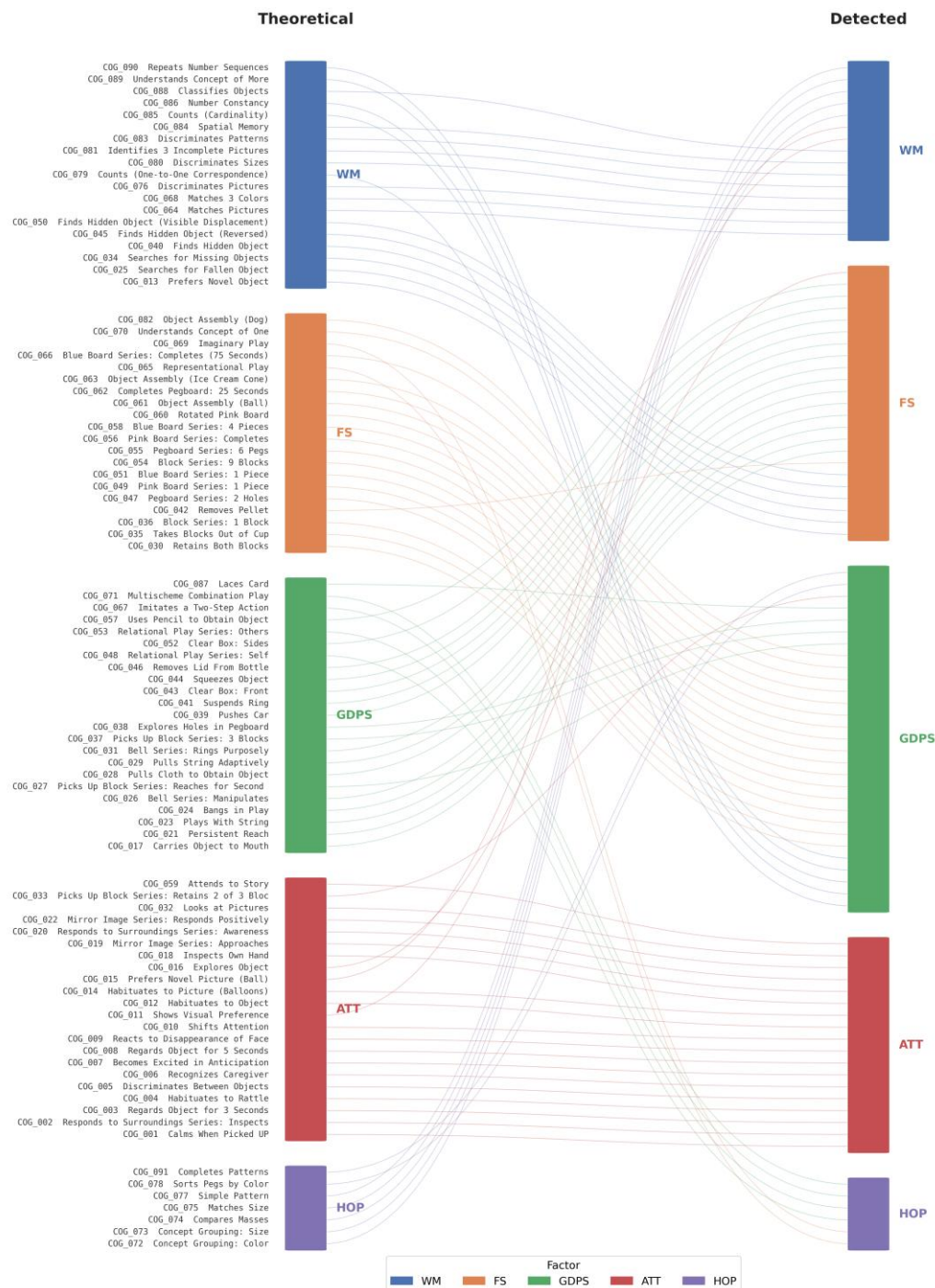
Centroid similarity heatmap for all 52 unpaired items.

Figure S3. Three-way Bayley-4 alluvial diagram



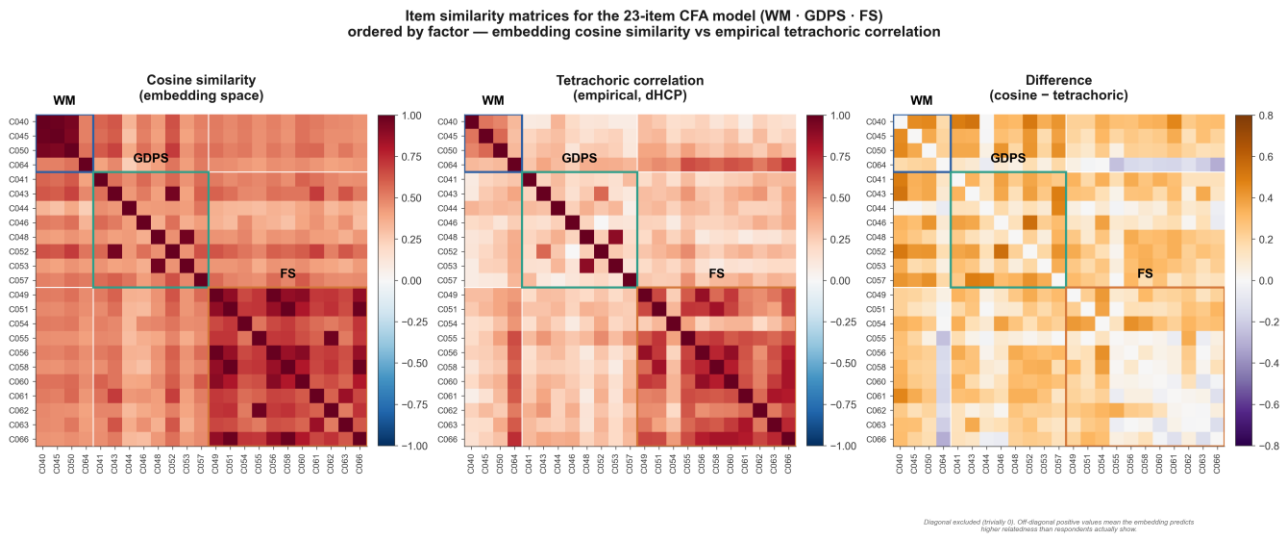
Three-way Bayley-4 alluvial diagram: item flow from the theoretical (Aylward et al. 2022), detected (unsupervised consensus k-means), and empirical (standardisation-sample) domain structures.

Figure S4. Bayley-III alluvial diagram



Bayley-III alluvial diagram: item flow from the theoretical (translated, Step 1) to the detected (bottom-up NLP, Step 3) domain structure, all 91 items.

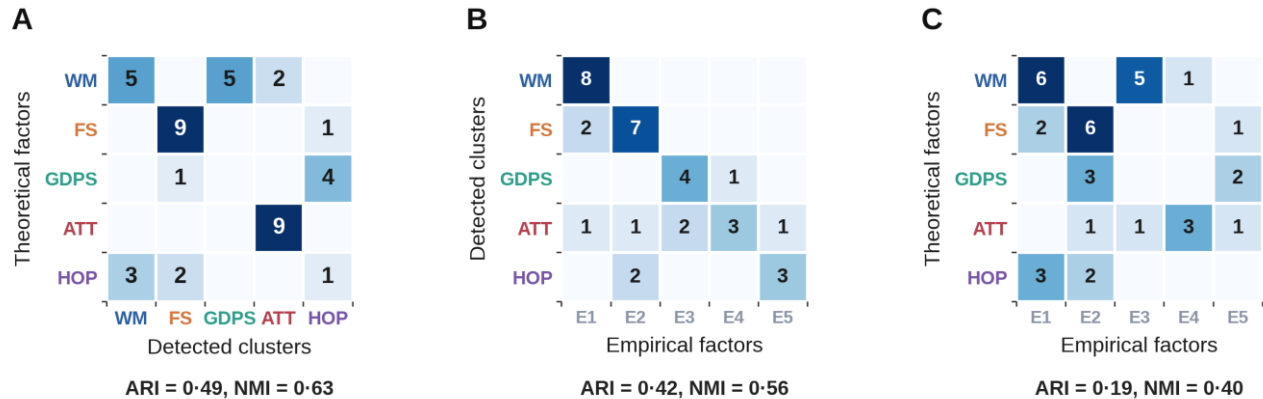
Figure S5. Embedding cosine similarity vs tetrachoric correlation, 23-item CFA model



Item similarity matrices for the 23-item CFA model (WM · GDPS · FS): embedding cosine similarity vs empirical tetrachoric correlation, dHCP sample.

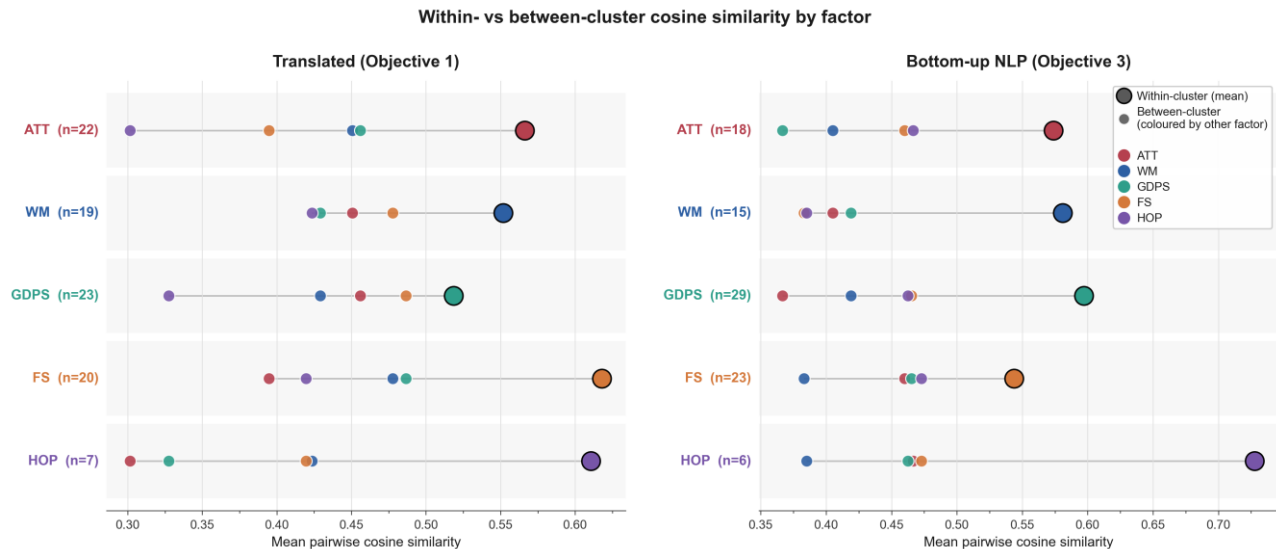
Figure S6. Three-way correspondence of the Bayley-4 domain structures

Three-way correspondence: theoretical, detected, and empirical Bayley-4 domain structures



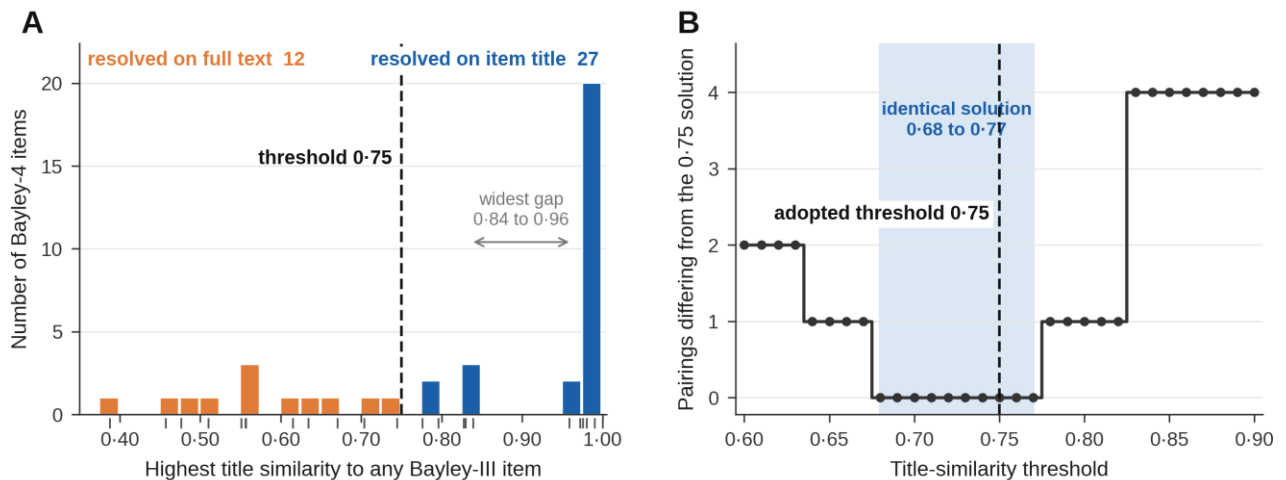
Three-way correspondence between theoretical, detected, and empirical Bayley-4 domain structures.

Figure S7. Within- vs between-cluster cosine similarity by factor



Within- vs between-cluster cosine similarity by factor, theoretical (Step 1) and bottom-up NLP (Step 3) solutions.

Figure S8. Sensitivity of the cascade pairing to the title-similarity threshold



Why cascade item pairing accepts a title match at 0.75. (A) Best title similarity for each of the 39 Bayley-4 cognitive items: 27 clear the threshold and pair on title, 12 fall through to full text. The widest gap is 0.84 to 0.96, so 0.75 marks a sparse region rather than the emptiest one. (B) Pairings differing from the adopted solution across thresholds 0.60 to 0.90. The solution is unchanged throughout 0.68 to 0.77. From 0.78 correct matches between reworded titles are rejected (Grouping: Size with Concept Grouping: Size); from 0.83 four pairings differ; below 0.68 spurious title matches are admitted, two by 0.63. 0.75 is therefore supported by the stability of the solution, not by a clean split in the distribution.

Appendix references

Numbered separately from the references of the main text; cited by superscript in the note to Table S4.

- 1 Cox JL, Holden JM, Sagovsky R. Detection of postnatal depression: development of the 10-item Edinburgh Postnatal Depression Scale. *Br J Psychiatry* 1987; 150: 782–86.
- 2 Arnold DS, O’Leary SG, Wolff LS, Acker MM. The Parenting Scale: a measure of dysfunctional parenting in discipline situations. *Psychol Assess* 1993; 5: 137–44.
- 3 Wolke D, Jaekel J, Hall J, Baumann N. Effects of sensitive parenting on the academic resilience of very preterm and very low birth weight adolescents. *J Adolesc Health* 2013; 53: 642–47.